FAIZ HUSSAIN

MLOps Engineer | DevOps Engineer
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PROFESSIONAL SUMMARY

MLOps and DevOps Engineer with 2+ years of experience automating AWS infrastructure and CI/CD delivery, now specialised in deploying and operating machine learning systems in production. Skilled in Terraform, Kubernetes, Docker, Python, and GitOps, with hands-on delivery of end-to-end ML pipelines covering training and containerised serving. Currently completing an MSc in Data Science at the University of Naples Federico II.

TECHNICAL SKILLS

Cloud: AWS (EC2, EKS, ECS, S3, Lambda, VPC, ECR, SageMaker, IAM)
MLOps: MLflow, Kubeflow, SageMaker Pipelines, DVC, FastAPI
Infrastructure as Code: Terraform, Helm, Ansible
Containers and Orchestration: Docker, Kubernetes (EKS)
CI/CD: GitHub Actions, GitLab CI, Jenkins, AWS CodePipeline
Monitoring: Prometheus, Grafana, CloudWatch
Programming: Python, SQL
Machine learning and Data Frameworks: scikit-learn, PyTorch, Pandas, Apache Spark, PostgreSQL

EDUCATION

MSc in Data Science
University of Naples Federico II, Italy | 2023 – 2026(Sept)

- Relevant coursework: Machine Learning, Statistical Modelling, Big Data (Spark), Databases, Cloud Computing.

BTech in Engineering
Dayananda Sagar University, India | 2017 - 2021

PROFESSIONAL EXPERIENCE

DevOps Engineer (Senior Associate, Information Technology) | NTT DATA
Bangalore, India | Sep 2021 - Aug 2023

- Built and managed AWS infrastructure using EC2, ECS, VPC, and S3 for production workloads.
- Automated infrastructure provisioning with Terraform, increasing deployment speed by 45%.
- Developed CI/CD pipelines in Jenkins and AWS CodePipeline, reducing release time from hours to minutes.
- Deployed and maintained containerised microservices on Kubernetes (EKS) using Docker and Helm.
- Configured CloudWatch monitoring, logging, and alerting to improve system reliability.
- Reduced AWS infrastructure costs by 20% through right-sizing and resource optimisation.
- Designed high-availability architecture using load balancing and auto-scaling.
- Worked in Agile sprints with developers and QA to deliver releases on schedule.

KEY PROJECTS

1. End-to-End MLOps Platform on AWS
Terraform | Kubernetes (EKS) | MLflow | Airflow | Argo CD | GitHub Actions | Python

- Provisioned the full platform with Terraform: EKS cluster, S3 artifact store, RDS metadata database, and IAM roles.
- Automated the training pipeline in Airflow, covering data validation, feature generation, training, and evaluation.
- Tracked experiments and versioned models in MLflow with an S3-backed model registry.
- Promoted approved models to staging and production automatically using Argo CD (GitOps).
- Result: fully reproducible training-to-deployment workflow with one-command rollback and no manual steps.

2. Real-Time Model Serving API with Drift Monitoring
FastAPI | Docker | Kubernetes | Prometheus | Grafana | Evidently AI

- Built a prediction API with FastAPI and packaged it as a lightweight multi-stage Docker image.
- Deployed on Kubernetes with Horizontal Pod Autoscaling to handle variable traffic.
- Exposed Prometheus metrics for latency, throughput, error rate, and prediction distribution.
- Detected data drift with Evidently AI and triggered automatic retraining via a webhook.

- Result: sub-100ms p95 latency with Grafana dashboards and automated drift alerting.

3. Containerised LLM Application on Kubernetes

Docker | Kubernetes | Terraform | vLLM | PostgreSQL (pgvector) | GitHub Actions

- Deployed an open-source LLM inference server on Terraform-provisioned Kubernetes nodes.
- Built a retrieval-augmented generation API backed by a pgvector document database.
- Automated build, test, and rollout of every component through GitHub Actions.
- Tracked token usage, latency, and cost per request in Grafana for capacity planning.
- Result: reproducible LLM serving stack with autoscaling and clear cost visibility.

CERTIFICATIONS

- AWS Certified DevOps Engineer, Professional
- AWS Certified Solutions Architect, Associate
- AWS Certified Cloud Practitioner

LANGUAGES

English: Professional (C1) | Italian: Elementary (A2, improving)